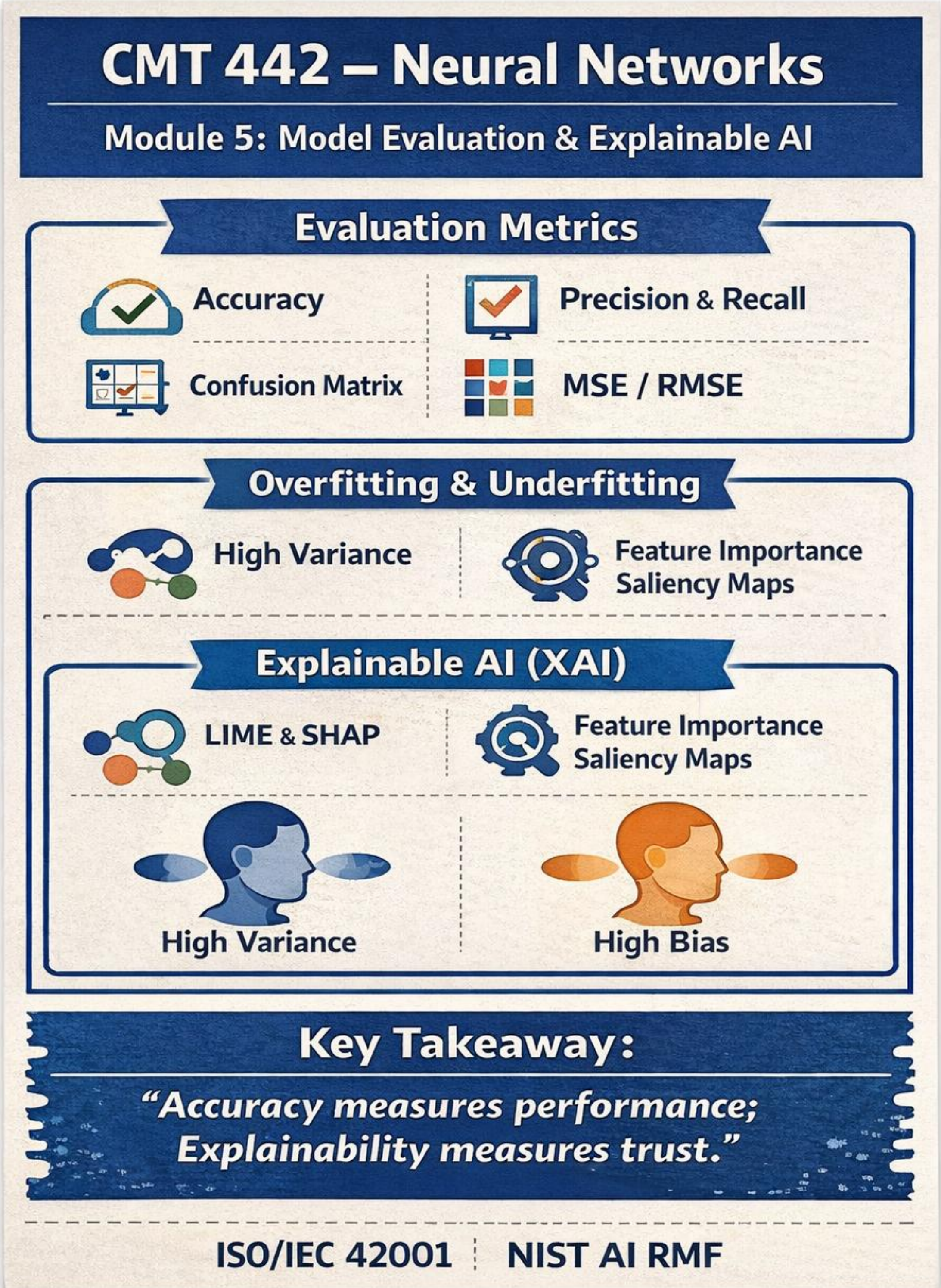
CMT 442 – Neural Networks

Module 5: Model Evaluation and Explainable AI

Duration: 1 Hour
Standards Alignment: ISO/IEC 42001 (Performance Evaluation), NIST AI RMF – Measure


CMT 442 – Neural Networks
Module 5: Model Evaluation & Explainable AI
Evaluation Metrics
Accuracy
Precision & Recall
Confusion Matrix
MSE / RMSE
Overfitting & Underfitting
High Variance
Feature Importance
Saliency Maps
Explainable AI (XAI)
LIME & SHAP
Feature Importance
Saliency Maps
High Variance
High Bias
Key Takeaway:
“Accuracy measures performance;
Explainability measures trust.”
ISO/IEC 42001
NIST AI RMF

Lecture Objectives

By the end of this lecture, students should be able to:

- Evaluate neural network performance using appropriate metrics
- Identify and mitigate overfitting and underfitting
- Explain AI model decisions using XAI techniques
- Link explainability to AI governance and accountability

Lecture Breakdown

0 – 10 min | Why Model Evaluation Matters

Key Concepts

- Difference between *training success* and *real-world performance*
- Why "high accuracy" can still mean **high risk**
- AI failures due to poor evaluation (bias, drift, opacity)

Instructor Emphasis

"If you cannot explain a model, you cannot govern it."

Prompt

- Why would a regulator reject a highly accurate AI model?

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10 – 25 min | Evaluation Metrics in Neural Networks

Classification Metrics

- Accuracy (and its limitations)
- Precision, Recall, F1-Score
- Confusion Matrix

Regression Metrics

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Advanced Concepts

- ROC curves & AUC
- Cost-sensitive evaluation

Example

- Medical diagnosis NN: false negatives vs false positives

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25 – 35 min | Overfitting, Underfitting & Generalization

Core Concepts

- Bias–variance trade-off
- Training vs validation vs test sets
- Data leakage risks

Mitigation Techniques

- Regularization (L1, L2)
- Dropout
- Early stopping
- Cross-validation

Mini Exercise

- Identify overfitting symptoms from learning curves

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35 – 50 min | Explainable AI (XAI)

Why Explainability Matters

- Trust
- Accountability
- Legal and ethical compliance
- ISO/IEC 42001 performance transparency

XAI Techniques

- Model-agnostic:
    - LIME
    - SHAP
- Model-specific:
    - Feature importance
    - Saliency maps (CNNs)

Case Study

- Loan approval neural network: explain rejection reasons

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50 – 60 min | Governance, Risk & Wrap-Up

Governance Link

- Explainability as an audit requirement
- Evidence for AI oversight committees
- Continuous performance monitoring

Discussion

- Should black-box AI be allowed in critical systems?

Key Takeaway

"Evaluation measures accuracy; explainability measures trust."

Module 5: Model Evaluation & Explainable AI

(10 CAT-style Scenario Questions)

1. A neural network used for loan approvals shows **96% accuracy**, but customer complaints are rising.
   a) Identify two evaluation metrics you would review beyond accuracy.
   b) Explain what risk the bank is currently exposed to.

2. During training, your model performs extremely well on training data but poorly on validation data.
   a) Name the issue.
   b) Propose two mitigation techniques and justify them.

3. A healthcare AI model misclassifies rare diseases.
   a) Which metric best captures this risk?
   b) Why would accuracy be misleading in this case?

4. Regulators request an explanation of why a patient was flagged as "high risk" by a neural network.
   a) Identify one suitable XAI technique.
   b) Explain how it improves accountability.

5. Two models have identical accuracy, but one has higher variance.
   a) Which model would you deploy in production?
   b) Defend your choice from a governance perspective.

6. A CNN shows unstable performance after deployment due to new data patterns.
   a) What evaluation weakness does this indicate?
   b) How would you address it?

7. A financial institution refuses to deploy a black-box model despite superior performance.
   Explain this decision using ISO/IEC 42001 principles.

8. A data scientist removes explainability tools to improve performance speed.
   a) Identify the governance risk.
   b) Recommend a balanced solution.
9. A confusion matrix shows high false negatives in fraud detection.
   Explain the business impact and corrective action.
10. Explain why **explainability is a performance requirement**, not just an ethical preference.